Chapter 15

Agriculture — II: Food Crops

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

.... crops are associated with the south-west monsoon.

1. Rabi
2. Kharif
3. Zayad
4. None of the above.

Answer

Kharif

Question 2

Which of the following is not a kharif crop?

1. Sugarcane
2. Cotton
3. Rice
4. Wheat.

Answer

Wheat

Question 3

The kharif crops are sown in June and July and harvested in ....

1. September and October
2. October and November

3. March and April
4. August and September

Answer

September and October

Question 4

Wheat and barley are associated with which agricultural season?

1. Kharif
2. Zaid
3. Rabi
4. None of the above.

Answer

Rabi

Question 5

The crops which are raised throughout the year are known as ....

1. Kharif
2. Rabi
3. Zaid
4. Baid

Answer

Zaid

Question 6

Indian agriculture is largely dominated by .... crops.

1. food
2. fibre
3. cash
4. spices

Answer

food

Question 7

.... are all kinds of grass-like plants, which have starchy edible seeds.

1. Cereals
2. Spices
3. Beverages
4. Oilseeds

Answer

Cereals

Question 8

India ranks .... in the production of cereals in the world.

1. first
2. second
3. third
4. fourth

Answer

third

Question 9

India is the .... largest producer of rice in the world.

1. fifth
2. fourth
3. third
4. second

Answer

second

Question 10

Rice grows best in ...., ... areas.

1. warm, humid
2. cold, dry
3. hot, rainy
4. hot, dry

Answer

warm, humid

Question 11

Rice thrives in the .... soils grown along the river banks.

1. laterite
2. red
3. black
4. alluvial

Answer

alluvial

Question 12

In the dry method of rice cultivation, seeds are sown in rows with drills in area of ....

1. low rainfall
2. heavy rainfall
3. moderate rainfall
4. scanty rainfall

Answer

heavy rainfall

Answer the following questions

Question 1(a)

Name the three cropping seasons in India.

Answer

The three cropping seasons in India are-

1. Kharif
2. Rabi
3. Zaid

Question 1(b)

Which is the most important method of rice cultivation in India? Why?

Answer

Japanese method of rice cultivation is highly adapted in India because the method includes the use of high yielding varieties of seeds called Japonica. Initial raising of the seedling is done in the nursery and is protected from initial infections and then transferred to fields and planted in rows to make weeding easy.

This method ensures higher yield.

Question 1(c)

Name the two states where rice is grown as a cash crop. Why?

Answer

In Punjab and West Bengal, the farmers cultivate rice as a cash or commercial crop. Rice is grown in almost all states of India except Gujarat and Rajasthan. Rice can feed more people compared to same amount of any other food grains.

Question 1(d)

Give three points of difference between upland and lowland rice.

Answer

Upland rice	Lowland rice
It is grown on mountainous regions.	It is grown in low-lying areas.
It is sown in March-April and harvested in September-October.	It is sown in June and harvested in October.
The entire crop is used locally.	The produce of rice is used for local consumption as to other regions.

Question 2(a)

Why are cereals referred to as 'staff of life'?

Answer

Cereals are referred to as 'staff of life' because of their simple form of cultivation and their high nutritional value. They have formed the basic diet of mankind.

Question 2(b)

State two geographical conditions suitable for the cultivation of rice?

Answer

Two geographical conditions suitable for the cultivation of rice are-

1. Rice grows best in warm and humid areas. The critical mean temperature for flowering and fertilisation ranges from 16°C to 20°C whereas during ripening, the range is from 18°C to 32°C.
2. Rice requires good rainfall in the range of 150 to 300 cm.

Question 2(c)

Give a geographical reason for each of the following:

(i) Wheat cultivation is confined to the northern parts of the country.

(ii) Punjab is the leading producer of wheat in India.

(iii) Government of India is encouraging cultivation of pulses.

Answer

(i) Wheat cultivation is confined to the northern parts of the country because it grows best in cool, moist climate and ripens in warm, dry climate. Also, the winter rain occurring in northern India is beneficial for the crop.

(ii) Punjab is the leading producer of wheat in India because the climate of Punjab is well suited for the cultivation of wheat. During winters, the temperature of the region remains between 10°C to 25°C. Punjab also receives light showers during the winter months from the western disturbances. Such a climate is suited for wheat cultivation.

(iii) Government of India is encouraging cultivation of pulses because pulses being leguminous crops, fix atmospheric nitrogen in the soil and increase the natural fertility of soil. Also, pulses form an important part of the Indian diet, especially for those who consume starchy vegetation.

Question 2(d)

(i) Name the state that produces the highest quantity of rice in India.

(ii) State two advantages of growing rice on lowlands.

Answer

(i) West Bengal produces the highest quantity of rice in India.

(ii) Two advantages of growing rice on lowlands are-

1. It is easy to irrigate rice crop and water can be easily accumulated on low lands.
2. Soil in the lowlands is more fertile as compared to uplands.

Question 3(a)

What kind of soil is needed for the cultivation of wheat?

Answer

Wheat grows best in well drained loams and clay loams.

Question 3(b)

State two geographical conditions necessary for the growth of wheat in India.

Answer

Two geographical conditions necessary for the growth of wheat in India are-

1. For the cultivation of wheat, temperature in the range of 10-15°C is suitable for sowing and 20-25°C during harvest.
2. About 80 cm of annual rainfall is ideal for wheat cultivation.

Question 3(c)

(i) Name two states that grow wheat extensively.

(ii) What climatic features have helped these states in this respect?

Answer

(i) Punjab and Uttar Pradesh grow wheat extensively.

(ii) Climatic features that have helped these states in this respect are-

1. Cool and moist climate for sowing and warm, dry climate for ripening. Ideal temperature ranges from 10-15°C at the time of sowing and 20-25°C during harvesting.
2. Annual rainfall of about 80 cm. Winter rain caused due to western disturbances are beneficial for the crop.

Question 3(d)

Give three differences between the climatic conditions needed for wheat and rice cultivation.

Answer

Climatic Conditions	Wheat	R
Temperature during sowing	10 to 15°C	16-20°C
Temperature during ripening/harvest	20-25°C	18-32°C
Rainfall	About 80 cm	About 150 - 300

Question 4(a)

What is meant by "transplantation"? State two of its advantages.

Answer

Transplantation is a method of rice cultivation common in deltaic and flood plain regions. The seedlings are first grown in nurseries and after four to five weeks when saplings attain 25-30 cm of height, they are transplanted into prepared rice fields in groups of four to six at a distance of 30-45 cm. In the beginning, the field is flooded with a 2-3 cm deep water. Subsequently, the depth of water level is increased to 4-6 cm till the crop matures.

Two advantages of transplantation method are-

1. Only the healthy plants are picked for resowing in the field and unhealthy plants are discarded.
2. Weeds are removed while resowing.

Question 4(b)

How does the cultivation of pulses usually help in restoration of fertility of the soil?

Answer

Pulses being leguminous crops, fix atmospheric nitrogen in the soil and increase the natural fertility of soil. Hence, pulses are usually rotated with other crops to maintain or restore soil fertility.

Question 4(c)

(i) Why are pulses grown as rotational crops?

(ii) Explain why India is the largest consumer of pulses.

Answer

(i) Pulses are grown as rotational crops as pulses being leguminous crops, fix atmospheric nitrogen in the soil and increase the natural fertility of soil.

(ii) India is the largest consumer of pulses because pulses form a very important part of the Indian diet, especially for those who consume starchy vegetarian diet. This is because pulses provide vegetable protein.

Question 4(d)

State three methods of growing rice.

Answer

Three methods of growing rice are-

1. The dry method of cultivation
2. The puddled or wet method
3. Transplanting method

Question 5(a)

What are millets?

Answer

The term 'millets' refers to a number of inferior grains like jowar, bajra and ragi, which serve as food grains for the poorer sections of the society.

Question 5(b)

Why are millets referred to as 'food grains of the poor'?

Answer

Millets are known as 'food grains of the poor' because these are coarse grains. Further, these do not require adequate water and can be grown in infertile soil owing to its rocky or sandy character.

Question 5(c)

Why are millets known as dry crops?

Answer

Millets are known as dry crops because they have a very short growing season and they can be grown under dry and high temperature conditions. Also, millets do not need rain or water and can survive in drought and other extreme conditions.

Question 5(d)

In what way are the millets different from rice?

Answer

Rice requires high temperature with adequate water for irrigation while millets can be grown on infertile soil and does not require much water for growth.

Millets have a higher nutritional value than rice.

Question 6(a)

In which region is ragi grown in India? Why?

Answer

Karnataka is the leading producer of ragi in the country followed by Tamil Nadu, Uttarakhand, Maharashtra, Andhra Pradesh, Odisha, Jharkhand, Rajasthan and Gujarat.

Ragi is grown in these states as the red, light black and sandy loams in Karnataka and Tamil Nadu and the well drained alluvial loams of Uttarakhand, Jharkhand and Gujarat are suitable for the cultivation of ragi. Also, the climatic conditions favour the cultivation of this crop.

Question 6(b)

In which part of the year is wheat grown in India? Why?

Answer

Wheat is a rabi crop and mostly grown during the cold weather season. It is usually sown in October and continues till the mid of November. It is harvested by the end of January in the south, and by March-April in the north.

It is so because wheat grows best in cool moist climate and ripens in warm, dry climate.

Question 6(c)

Why is wheat not grown in the eastern and in the extreme southern parts of India? Name the state that is the largest producer of wheat.

Answer

Wheat is not grown in the eastern and in the extreme southern parts of India because the temperature required at the time of sowing wheat is 10-15°C which is not possible in these regions.

Also, wheat grows best in well drained loams and clay loam whereas the southern and eastern India have laterite and red soil.

Uttar Pradesh is the largest producer of wheat in India.

Question 6(d)

Which is a useful 'rotation crop'? Give reasons to support your answer.

Answer

Pulses are useful rotation crops as pulses being leguminous crops, fix atmospheric nitrogen in the soil and increase the natural fertility of soil.

Question 7

Study the picture given below and answer the following questions:

(a) Name the crop which is being planted. Give one benefit of this method of planting this crop.

(b) Name the other method of planting the crop. In which area is this method practised?

(c) Give a geographical reason for each of the following:

(i) Rice is not the main crop in the Deccan Plateau.

(ii) Punjab is the largest producer of rice despite deficient rainfall.

(iii) Wheat grows well in loamy soil.

(d) Mention the climatic conditions which favour the cultivation of the crop being planted.

Answer

(a) Rice is being planted. A benefit of this method of plantation is that this method gives higher yield.

(b) Another method of planting the crop is drilling method. It is practised in Peninsular India.

(c)

(i) Rice is not the main crop in the Deccan Plateau because the soil required for rice cultivation is clayey or loamy soil while deccan plateau has black soil. Also, rice requires adequate water for irrigation and this region falls in the rain shadow region of the western ghats.

(ii) Punjab is the largest producer of rice despite deficient rainfall because here, 97% of the rice area is irrigated and due to higher input of High Yielding Variety seeds, fertilisers and mechanisation the per hectare yield is the highest.

(iii) Wheat grows well in loamy soil because loamy soil is a mixture of sand, silt and clay which provides ideal nutrition for wheat. The fertile, well-drained loamy soil allows for rapid absorption of water and air by plant roots, which encourages growth of the wheat plant.

(d) The climatic conditions suitable for the cultivation of rice are-

1. Rice grows best in warm and humid areas. The critical mean temperature for flowering and fertilisation ranges from 16°C to 20°C whereas during ripening, the range is from 18°C to 32°C.
2. Rice requires good rainfall in the range of 150 to 300 cm.

Thinking Skills

Question 1

Indian agriculture is largely dominated by food crops. What is the reason for it? Give examples to support your answer.

Answer

Indian agriculture is largely dominated by food crops due to the following reasons:

1. Food Security — India has a large population, and ensuring food security is a top priority. Food crops such as wheat, rice, and millets are essential for meeting the country's dietary needs and preventing hunger.
2. Subsistence Agriculture — Many farmers practice subsistence agriculture, where the focus is on growing crops for personal consumption rather than commercial purposes.
3. Demand and Market Stability — There is a consistent and stable demand for food crops in the domestic market as food crops are essential staples in Indian cuisine.
4. Government Policies and Support — The Indian government has historically provided support, subsidies and incentives for food crop cultivation to ensure food self-sufficiency and stabilize prices in the market. This further encourages farmers to focus on growing food crops.

Some examples of cultivation of food crops are as follows:

1. Rice is a major food crop grown extensively in states like West Bengal, Punjab, and Tamil Nadu. It is a staple food for a large part of the Indian population.
2. Wheat is another significant food crop grown in regions like Punjab, Haryana, and Uttar Pradesh. It is an important staple in the northern parts of India.

3. Millets, such as bajra and jowar, are widely cultivated in states like Rajasthan, Maharashtra, and Karnataka. These crops are drought-resistant and serve as important food sources, particularly in arid and semi-arid regions.

Question 2

You live in a state which has a number of rivers, deltas and estuaries, good rainfall and heavy soils. Which cereal crop is ideal for growth in your state? During which season would this crop be sown and which method of cultivation would be used?

Answer

Rice crop is ideal for growth in my state.

Rice is a kharif crop. It is sown in the month of June-July and harvested in the month of September-October.

The puddled or wet method of cultivation would be used as we have assured supply of water for irrigation. In this method, the land is ploughed thoroughly and filled with three to five centimetres of standing water in the field. This water is maintained in the fields up to a depth of two to three centimetres till the seedlings are well established.

Question 3

Though India is an agricultural country, the cost of food is rising sharply. Give reasons to explain the causes of this rise in prices of food.

Answer

The rising cost of food in India can be attributed to several factors:

1. Increasing Population — The rising population in India leads to increased demand for food, putting pressure on the available supply and driving up prices.
2. Inflation and Economic Factors — Inflationary pressures, rising fuel prices, transportation costs, and input costs for farmers contribute to higher food prices.
3. Weather and Climate Change — Adverse weather events like droughts, floods and extreme temperatures affect crop yields and reduce the overall food supply, leading to price increases.
4. Supply Chain Issues — Inadequate infrastructure, storage facilities and transportation networks result in post-harvest losses and wastage, leading to higher prices.
5. Income Disparities — Income disparities affect food prices, as higher-income groups demand costlier food products, driving up prices.
6. Export and Import Policies — Trade policies and restrictions on agricultural commodities can create imbalances between supply and demand, impacting domestic food prices.
7. Global Market Influences — Global commodity prices, exchange rates, and geopolitical events have a trickle-down effect on domestic food prices in India.